
Translate Needs To Integration Requirements

Given an existing system landscape diagram, create an inventory of the systems and integration patterns.

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Business Scenario

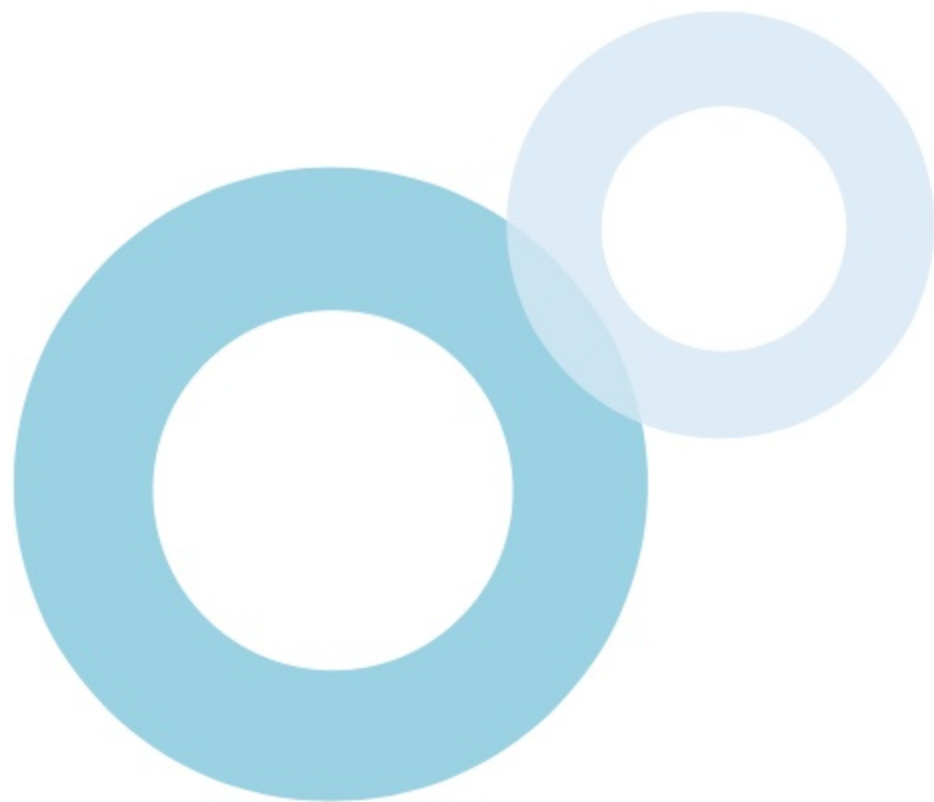


Requirements & Solutions



After studying this topic, you should be able to:

-  Identify the different types of Salesforce system architecture landscapes.
-  Identify the types of integration patterns that can be used in a system landscape.
-  Given a system architecture diagram, identify the systems and integration patterns used for integration.



Introduction

There are different types of Salesforce system architecture landscapes, namely, ground-to-cloud, cloud-to-ground, and cloud-to-cloud. In the **Cloud-to-Ground** architecture, data originates from Salesforce and is pushed into an on-premise system. In the **Ground-to-Cloud** architecture, data originates from the on-premise architecture and is pushed into Salesforce. In the **Cloud-to-Cloud** architecture, data is moved from one org to another. The integration pattern(s) used in a system landscape depend on the company's business requirements. The different integration patterns are **Remote Process Invocation - Request & Reply**, **Remote Process Invocation - Fire & Forget**, **Batch Data Synchronization**, **Remote Call-In**, **UI Update Based on Data Changes**, and **Data Virtualization**.



Overview

Cloud-to-Ground Architecture

In a cloud-to-ground system landscape, a transaction (message or data) is pushed from Salesforce into the on-premise infrastructure.

Ground-to-Cloud Architecture

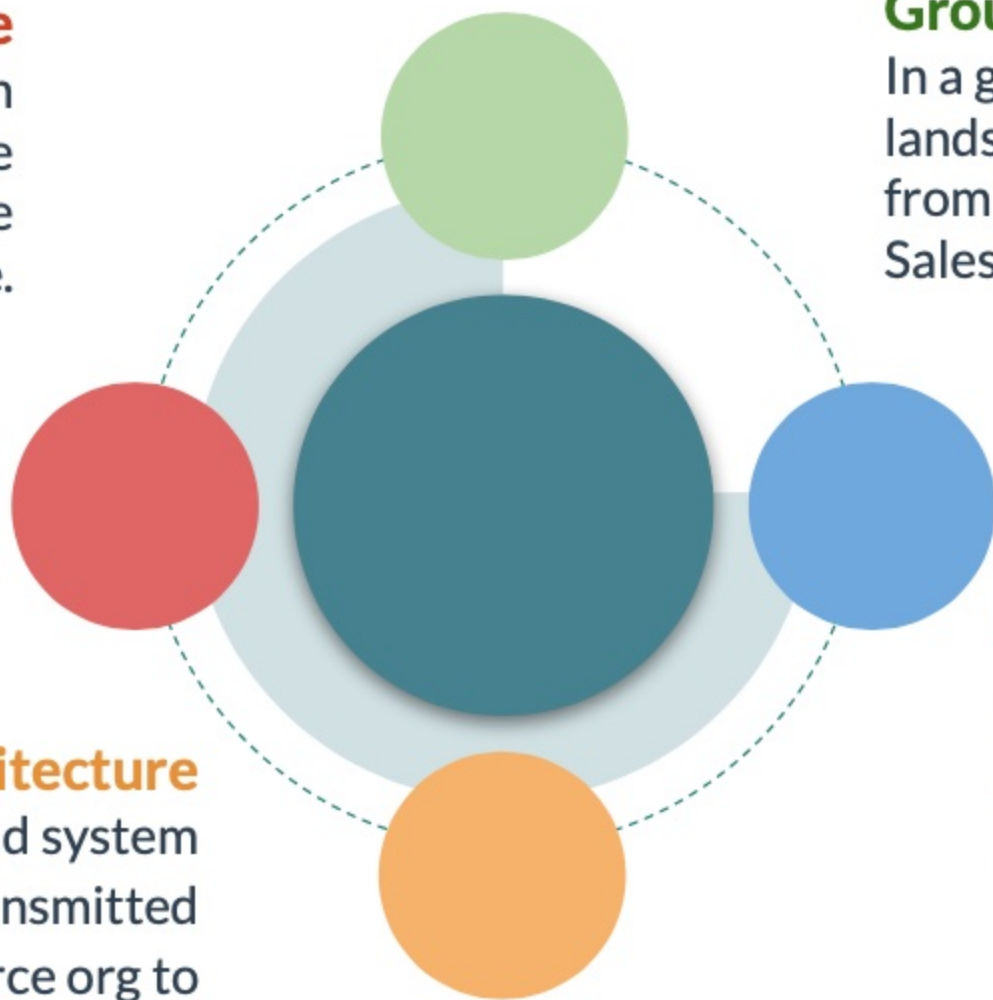
In a ground-to-cloud system landscape, data is pulled and pushed from the on-premise infrastructure to Salesforce.

Cloud-to-Cloud Architecture

In a cloud-to-cloud system landscape, data is transmitted from one Salesforce org to another.

Integration Patterns

The integration patterns used in a system landscape depend on the company's business requirements.



Types of System Landscapes

There are different types of **Salesforce system architecture landscapes**. Each system landscape consists of multiple systems or platforms.

❖ CLOUD-TO-GROUND

In a cloud-to-ground system landscape, a **transaction (message or data)** is pushed **from Salesforce into the on-premise infrastructure**.

❖ GROUND-TO-CLOUD

In a ground-to-cloud system landscape, **data** is **pulled and pushed** from the **on-premise infrastructure to Salesforce**.

❖ CLOUD-TO-CLOUD

In a cloud-to-cloud system landscape, **data** is **transmitted from one Salesforce org to another**.



Cloud-to-Ground Architecture

In the **Cloud-to-Ground architecture**, data originates from Salesforce. It may consist of various capabilities.

❖ SALESFORCE TO DMZ

A message that originates **from Salesforce can be relayed to a DMZ (demilitarized zone) service end-point**, which could be a firewall, a services gateway appliance, or reverse proxy.

❖ DMZ TO ON-PREMISE

The message **from the DMZ can be relayed into the trusted on-premise infrastructure**, usually destined for an **Enterprise Service Bus (ESB)**, which would handle any **transformation, mediation, and orchestration**.

❖ ESB TO SOA

The **ESB** may push the message into the **SOA (Service Oriented Architecture)** infrastructure, which comprises web services. Salesforce can become a consumer (and later a producer) of these SOA services.



Cloud-to-Ground Architecture

In the **Cloud-to-Ground architecture**, data originates from Salesforce. It may consist of various capabilities.

❖ DATABASE ACCESS

Salesforce can **read** data from an **on-premise database** in **real (or near-real) time**, which may be a standalone database or part of a more formal **Enterprise Data Warehouse (including an ODS – an operational data store)**.



Ground-to-Cloud Architecture

In the **Ground-to-Cloud architecture**, data originates from the on-premise architecture. It may consist of various capabilities.

❖ CALLS TO SALESFORCE

An **ESB** can handle all of the real-time calls into Salesforce. In the absence of an ESB, **each separate application** requiring access would handle the calls. However, it is better to use a **centralized integration middleware** using **OAuth** or **username-password authentication**.

❖ BATCH DESIGN

An **ETL tool** can be used to get **large volumes of data** in and out of Salesforce in batches. **Bulk API** can be used where possible.

❖ DATA REPLICATION & RESTORATION

Salesforce data can be off-loaded into a **replicated copy** in an **on-premise platform** to enable **restoration to an earlier state** later. The replications should be **processing diffs** and **never whole tables** to ensure scalability.



Ground-to-Cloud Architecture

In the **Ground-to-Cloud architecture**, data originates from the on-premise architecture. It may consist of various capabilities.

❖ DATABASE TO SALESFORCE

Data may be moved from **EDW (Enterprise Data Warehouse)** to **Salesforce** using an **ETL tool**.



Cloud-to-Cloud Architecture

In the **Cloud-to-Cloud architecture**, data is moved from one org to another. It may consist of various capabilities.

❖ SALESFORCE2SALESFORCE

Salesforce2Salesforce can be used to **integrate multiple Salesforce orgs**. Alternatively, **RESTful web services integration** can be used.

❖ POINT-TO-POINT INTEGRATION

A **middleware** can be used for **point-to-point integration** with other systems. The **use of Apex should be avoided**.

❖ CLOUD-BASED INTEGRATION

Cloud-based Integration-as-a-Service packages may be used for managing cloud-to-cloud use cases.



Cloud-to-Cloud Architecture

In the **Cloud-to-Cloud architecture**, data is moved from one org to another. It may consist of various capabilities.

❖ CLOUD SERVICE BUS

A **middleware** can be used to handle **service mediation, transformation, routing, and error handling** to other cloud-based end points.

❖ ESB INTEGRATIONS

A company may broker **all integrations through their ESB**, including **cloud-to-cloud** use cases. However, the cost of a highly resilient ESB may be extremely high.



Integration Patterns

The **integration patterns** used in an **architecture** depend on the company's **use cases**.

❖ REMOTE PROCESS INVOCATION

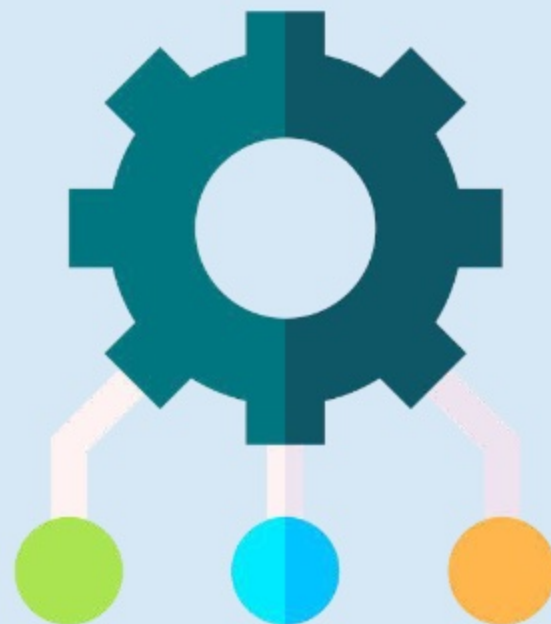
A **remote process in an on-premise system** can be invoked from Salesforce **synchronously or asynchronously**, depending on the business process requirement.

❖ BATCH DATA SYNCHRONIZATION

An **ETL tool** can be used to **synchronize data** between Salesforce and an on-premise system.

❖ REMOTE CALL-IN

A **remote process in an on-premise system** can perform an **API call** to Salesforce or **invoke an Apex REST or web service**.



Integration Patterns

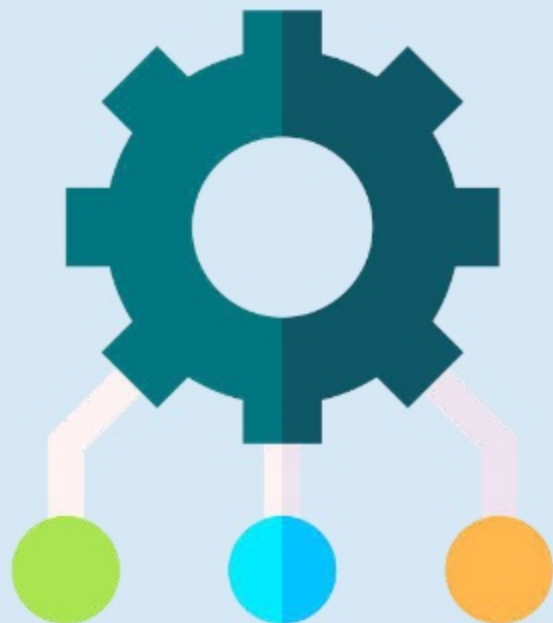
The **integration patterns** used in an **architecture** depend on the company's **use cases**.

❖ UI UPDATE

A **remote process in an on-premise system** can use the **Streaming API** to stream notifications to a Salesforce user interface.

❖ DATA VIRTUALIZATION

Data stored in an on-premise system can be exposed in Salesforce via **Salesforce Connect**.



Business Scenario

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Business Scenario

Read the business scenario below and determine the most appropriate solutions for the company's requirements.



Cosmic Furniture is a multinational company that sells furniture for homes, such as bedroom and living room furniture. It has its headquarters in the United States and has sold its products to more than 10 million customers.

The company uses Salesforce for managing accounts, contacts, leads, opportunities, and cases. Other enterprise systems, such as an enterprise resource planning (ERP) system, are used for operations such as order management and performance management. As a part of a new integration maintenance project, the solution architect of the company needs to analyze the existing system landscape based on specific business requirements and create an inventory of the systems and integration patterns used for integrating the systems with one another.

Requirements & Solutions

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Requirement 1

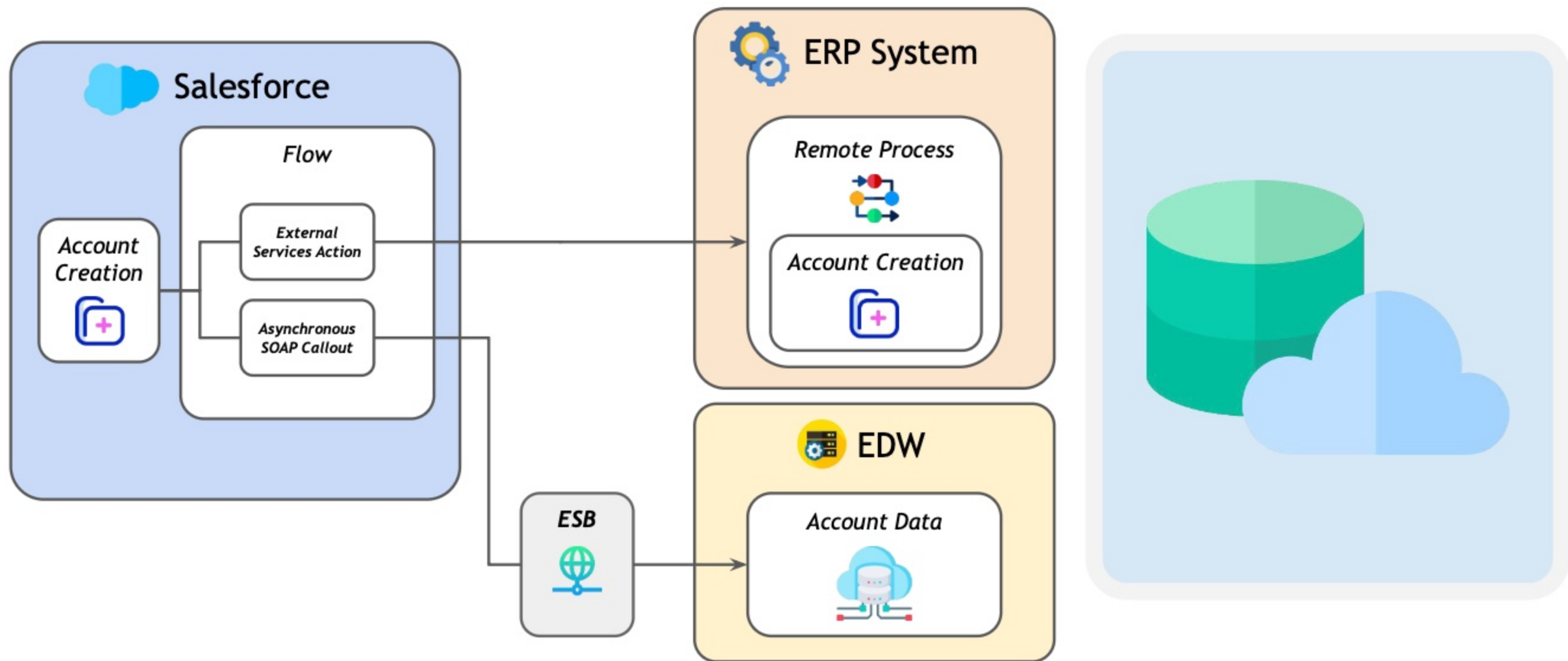


SCENARIO

Salesforce is the primary system used for managing accounts. However, when a new account is created, it is automatically inserted into the ERP system using a record-triggered flow that invokes an External Services action. A transaction ID that is included in the response from the ERP after the request is made should be saved into the Account record for reference. In addition, the flow also inserts the account into an EDW (Enterprise Data Warehouse) that is used for a Business Intelligence tool. An asynchronous Apex SOAP callout is used to route the account data to the data warehouse via an ESB.



Requirement 1



Solution 1



SOLUTION

Systems & Integration Patterns

- 1) **Salesforce to ERP (External Services):**
Remote Process Invocation (Request & Reply)
- 2) **Salesforce to EDW (Async SOAP Callout):**
Remote Process Invocation (Fire & Forget)



Requirement 2

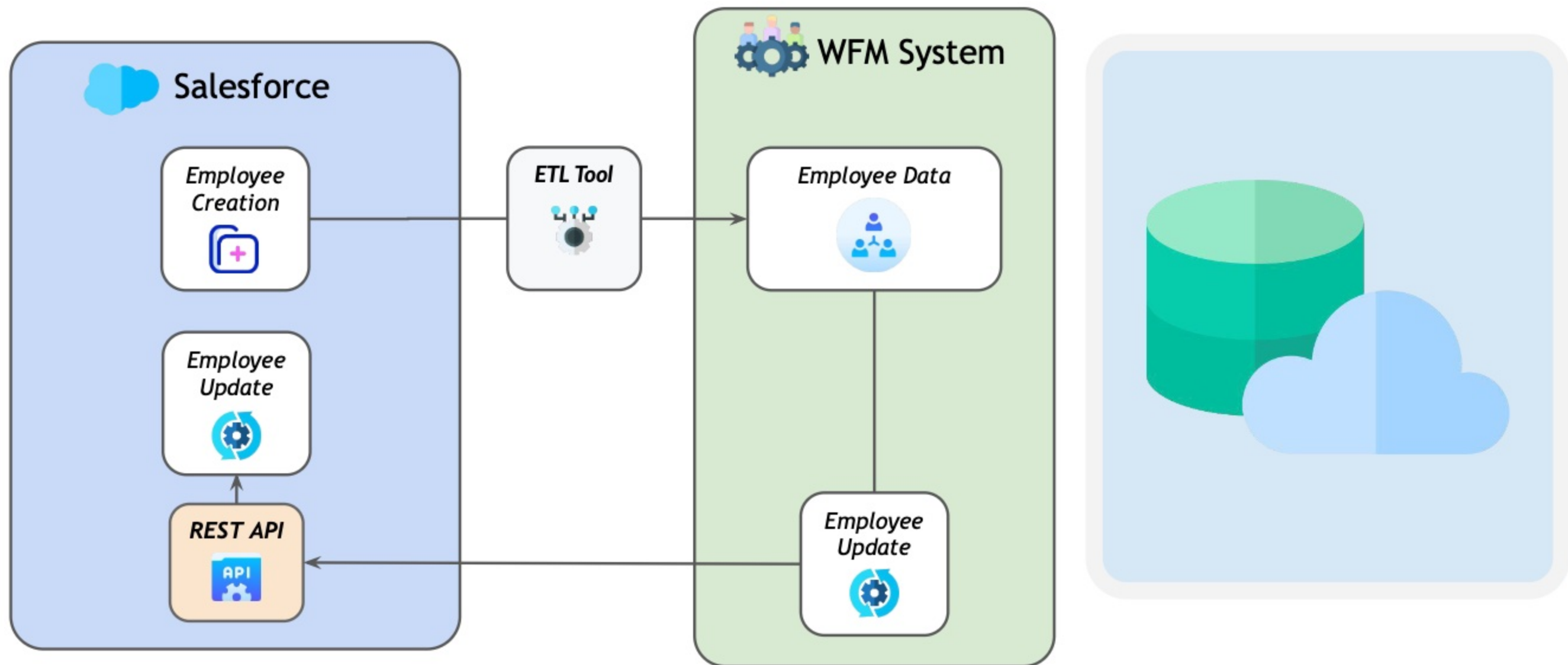


SCENARIO

The sales department uses Salesforce to manage information about sales employees using records of a custom object named 'Employee__c'. Since the HR department of the company primarily uses the WFM system, all the employee records created on a particular day are loaded into the WFM system using an ETL tool at the end of the day. If an employee record is updated in the WFM system, data containing information about the update is sent to Salesforce via a synchronous REST API call.



Requirement 2



Solution 2



SOLUTION

Systems & Integration Patterns

- 1) **Salesforce to WFM (ETL Tool):**
Batch Data Synchronization
- 2) **WFM to Salesforce (REST API):**
Remote Call-In



Requirement 3

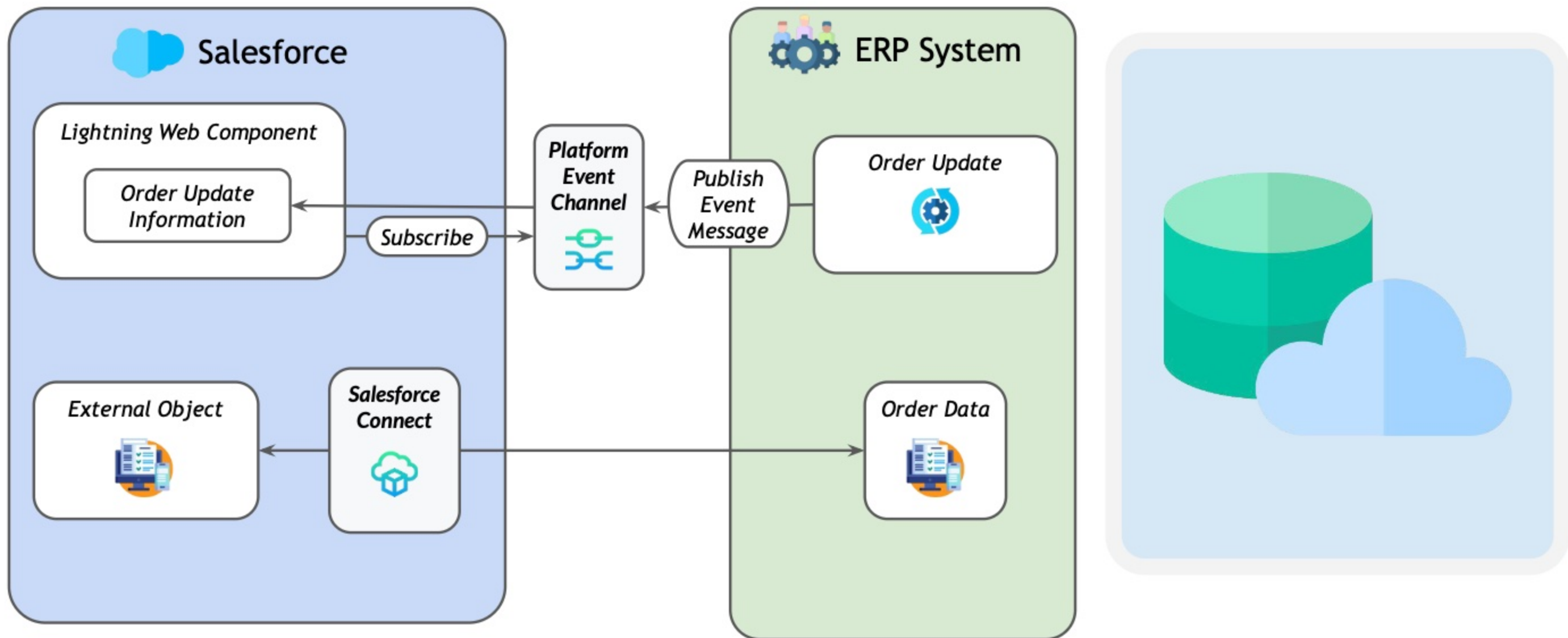


SCENARIO

An order placed by a customer goes through various stages in the ERP system, such as 'Created', 'Fulfilled', and 'Shipped'. When an order reaches a new stage, a platform event message is published using REST API. A Lightning web component used by sales users has subscribed to receive the platform event messages. When a new platform event message is received, the Lightning web component shows information about the order including its stage. Furthermore, all the order records in the ERP system are exposed in Salesforce via Salesforce Connect.



Requirement 3



Solution 3



SOLUTION

Systems & Integration Patterns

- 1) **ERP to Salesforce (Platform Event):**
UI Update Based on Data Changes
- 2) **Salesforce to ERP (Salesforce Connect):**
Data Virtualization



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[Integration Architecture for the Salesforce Platform](#)